25 Introduction to Bayesian Hierarchical Linear Models

We have studied a Bayesian simple linear regression model. We have also introduced some basic principles of hierarchical models. Now we'll combine elements to formulate Bayesian hierarchical linear regression models.

This handout corresponds to Chapters 15-17 of (Johnson et al. 2022). This handout only contains a brief introduction; see (Johnson et al. 2022) for a much more thorough analysis.

We'll consider the `cherry_blossom_sample` data in the `bayesrules` package which contains the `net` running times (in minutes) and `age` for participants in the annual 10-mile Cherry Blossom race held in Washington, D.C. Each runner in the sample is in their 50s or 60s.

```
library(bayesrules)

data(cherry_blossom_sample)
running <- cherry_blossom_sample |>
  select(runner, age, net) |>
  na.omit()

head(running)
```

```
# A tibble: 6 x 3
  runner   age   net
  <fct>  <int> <dbl>
1 1         53  84.0
2 1         54  74.3
3 1         55  75.2
4 1         56  74.2
5 1         57  74.2
6 2         52  82.7
```

1. Suppose we fit a Bayesian simple regression model of `net` on `age`. What do the slope and intercept parameters represent in context? What might be a reasonable prior? (You can assume that the `age` variable has been centered by subtracting the sample mean.)

2. Fit the model from the previous part (see the results in Section 17.1 of (Johnson et al. 2022)). Do the results seem reasonable? (Hint: consider the posterior distribution of the slope.)

3. Each of the runners in the sample has run the race in multiple years. What assumption of the simple linear regression model is violated?

4. Formulate a Bayesian hierarchical linear model which assumes the intercept (but not slope) varies by runner. What do the parameters represent in context?

5. Fit the model from the previous part (see the results in Section 17.2.4 of (Johnson et al. 2022)). What are some features of the analysis you would explore? Consider the posterior distribution of the slope associated with `age`; does the posterior distribution based on this hierarchical model seem more reasonable than the one based on the simple linear regression model?

6. Formulate a Bayesian hierarchical linear model which assumes both the intercept and the slope vary by runner. What do the parameters represent in context?

7. Fit the model from the previous part (see the results in Section 17.3.3 of (Johnson et al. 2022)). What are some features of the analysis you would explore?

8. Compare the two hierarchical regression models. What are some similarities? What are some differences?

25.1 Notes

```r
library(brms)
```

```
Loading required package: Rcpp

Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').


Attaching package: 'brms'

The following objects are masked from 'package:tidybayes':

    dstudent_t, pstudent_t, qstudent_t, rstudent_t

The following object is masked from 'package:stats':

    ar
```

```r
library(bayesplot)
```

```
This is bayesplot version 1.14.0

- Online documentation and vignettes at mc-stan.org/bayesplot

- bayesplot theme set to bayesplot::theme_default()

   * Does _not_ affect other ggplot2 plots

   * See ?bayesplot_theme_set for details on theme setting
```

```
Attaching package: 'bayesplot'

The following object is masked from 'package:brms':

    rhat
```

```
library(tidybayes)
```

25.1.1 Linear model

```
fit_lm <- brm(data = running, family = gaussian(), net ~ 1 + age)
```

```
Compiling Stan program...

Start sampling


SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 3.1e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.31 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.023 seconds (Warm-up)
```

```
Chain 1:                0.008 seconds (Sampling)
Chain 1:                0.031 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 3e-06 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.03 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.024 seconds (Warm-up)
Chain 2:                0.009 seconds (Sampling)
Chain 2:                0.033 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 5e-06 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.05 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
```

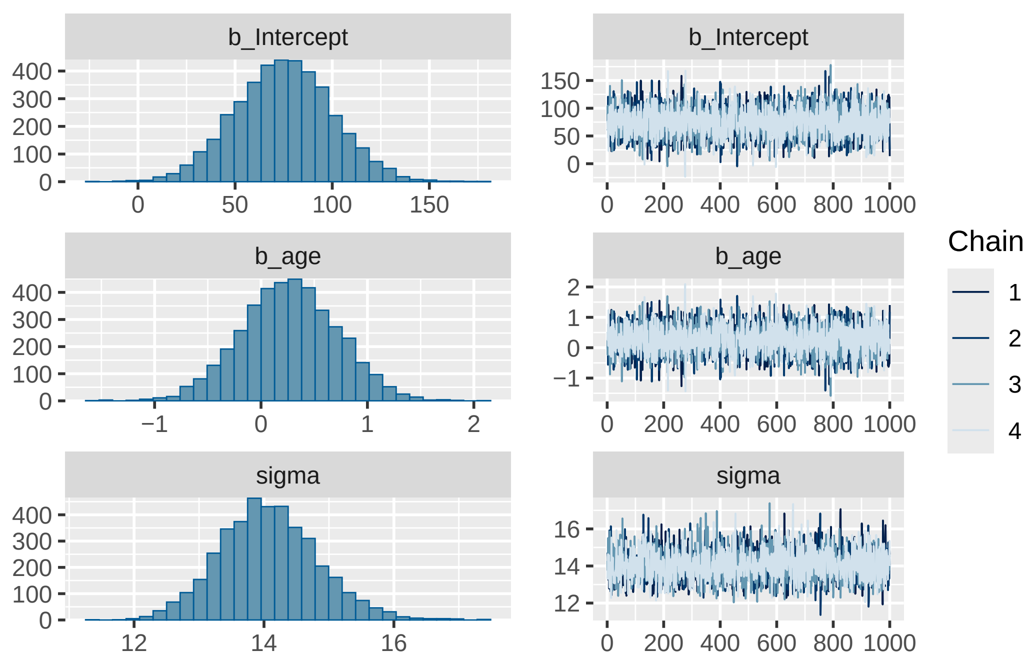
```
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.025 seconds (Warm-up)
Chain 3:                0.009 seconds (Sampling)
Chain 3:                0.034 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 1e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.1 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4:
Chain 4:  Elapsed Time: 0.022 seconds (Warm-up)
Chain 4:                0.011 seconds (Sampling)
Chain 4:                0.033 seconds (Total)
Chain 4:
```

```r
summary(fit_lm)
```

```
 Family: gaussian
  Links: mu = identity
Formula: net ~ 1 + age
   Data: running (Number of observations: 185)
```

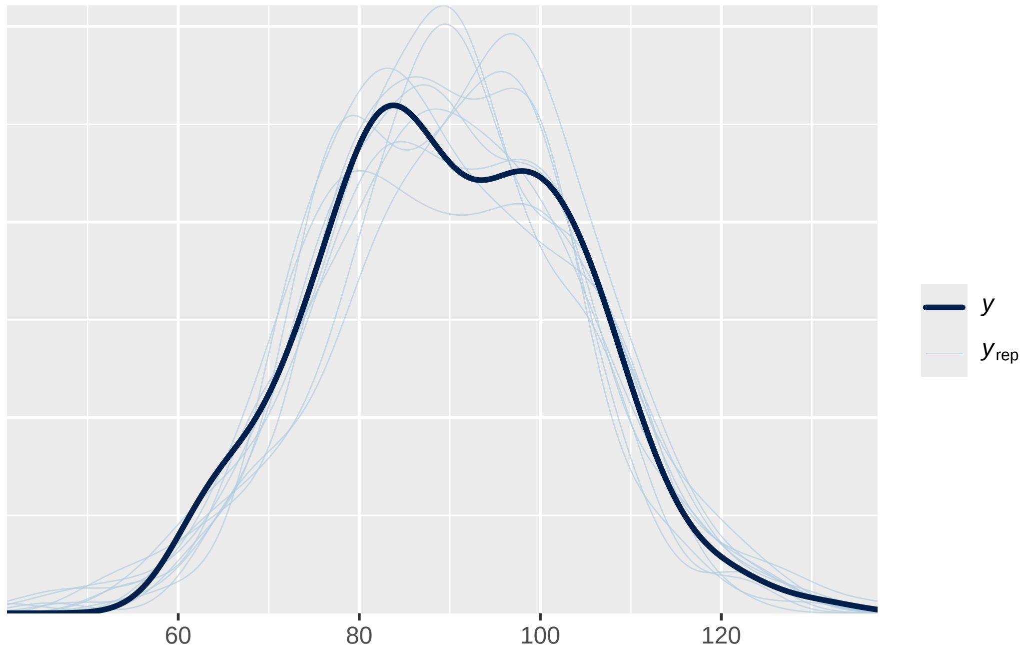
```
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    75.16     24.94    26.40   124.04 1.00     4648     2589
age           0.27      0.45    -0.62     1.15 1.00     4612     2572

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma    14.08      0.76    12.63    15.67 1.00     4149     2923

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit_lm)
```




```r
pp_check(fit_lm)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



25.1.2 Hierarchical model - random intercepts only

```
fit_hier1 <- brm(
  data = running,
  family = gaussian(),
  net ~ 1 + age + (1 | runner)
)
```

```
Compiling Stan program...

Start sampling


SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 7e-05 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 0.7 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
```

```
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 0.441 seconds (Warm-up)
Chain 1:                0.216 seconds (Sampling)
Chain 1:                0.657 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 1.6e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.16 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 0.447 seconds (Warm-up)
Chain 2:                0.227 seconds (Sampling)
Chain 2:                0.674 seconds (Total)
Chain 2:
```

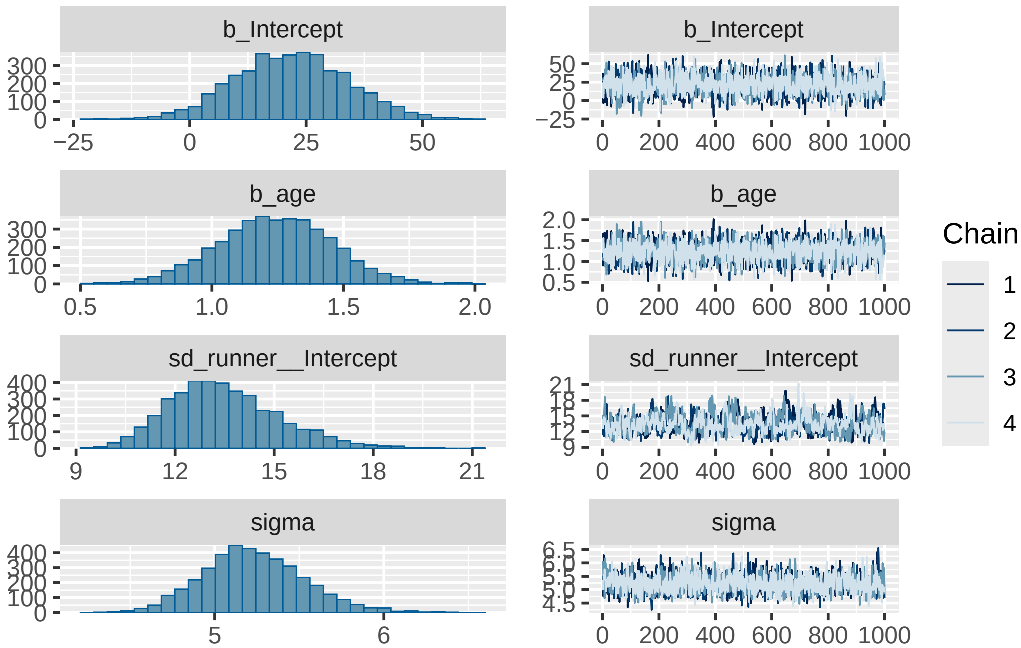
```
SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 1.6e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.16 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 0.35 seconds (Warm-up)
Chain 3:                0.198 seconds (Sampling)
Chain 3:                0.548 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 1.5e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.15 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
```

```
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4: 
Chain 4:  Elapsed Time: 0.691 seconds (Warm-up)
Chain 4:                0.173 seconds (Sampling)
Chain 4:                0.864 seconds (Total)
Chain 4: 

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and tai
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

```r
summary(fit_hier1)
```

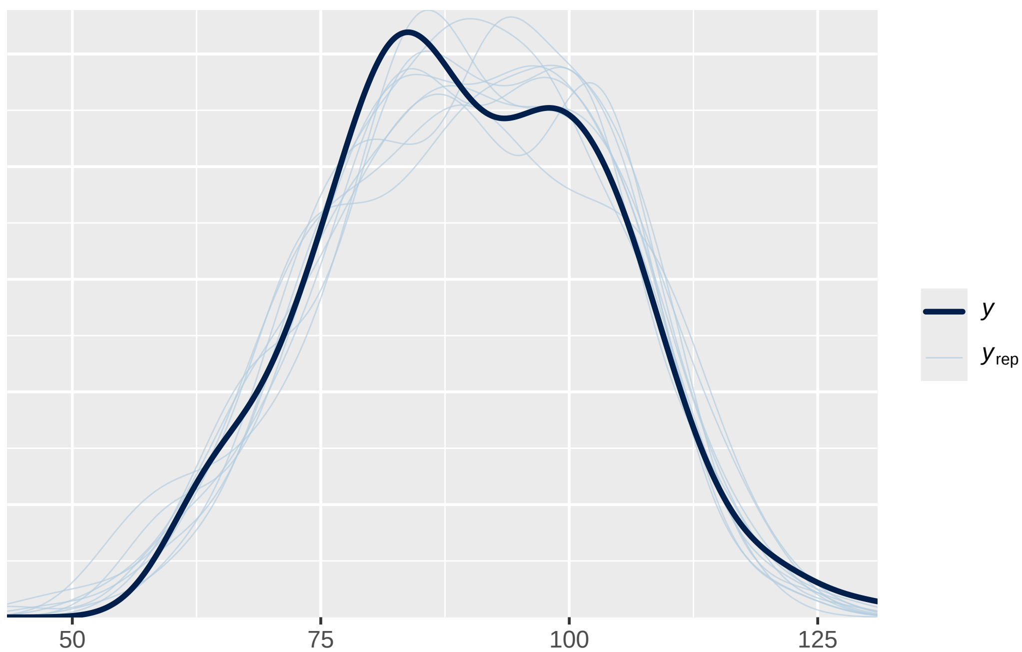
```
 Family: gaussian 
  Links: mu = identity 
Formula: net ~ 1 + age + (1 | runner) 
   Data: running (Number of observations: 185) 
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~runner (Number of levels: 36) 
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    13.49      1.65    10.70    17.17 1.01      353      598

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    21.86     12.30    -1.58    46.00 1.00     1695     2625
age           1.24      0.22     0.81     1.67 1.00     2206     2705

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     5.23      0.30     4.69     5.88 1.00     1850     2371

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit_hier1)
```




```r
pp_check(fit_hier1)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



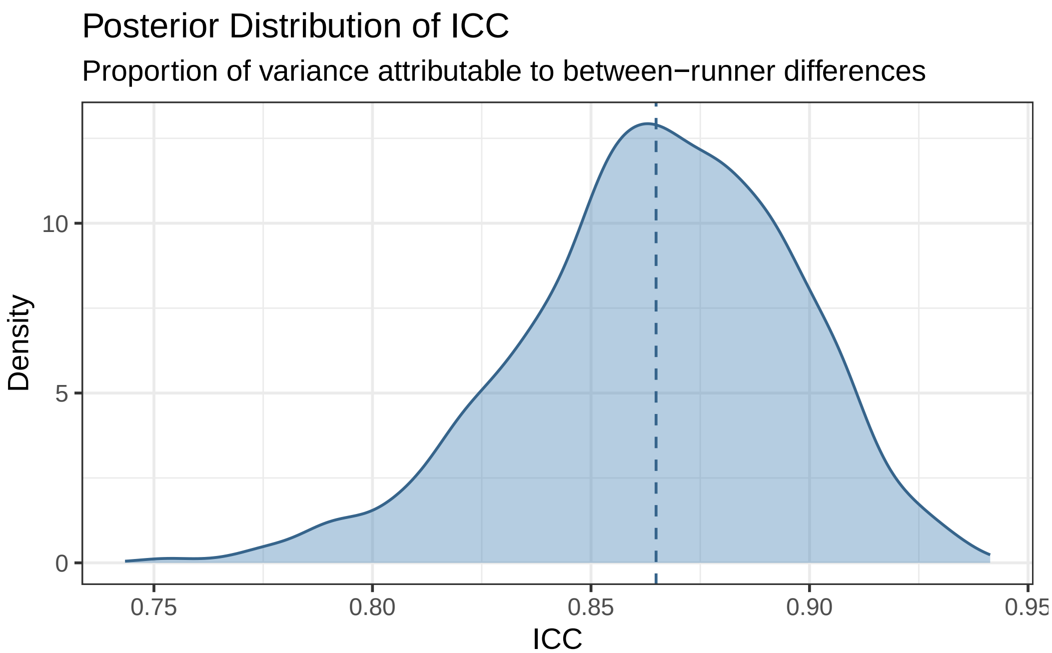

```
library(performance)
icc(fit_hier1)
```

```
# Intraclass Correlation Coefficient

    Adjusted ICC: 0.869
  Unadjusted ICC: 0.836
```

```
icc_draws <- fit_hier1 |>
  spread_draws(sd_runner__Intercept, sigma) |>
  mutate(
    var_between = sd_runner__Intercept^2,
    var_within = sigma^2,
    icc = var_between / (var_between + var_within)
  )
```

```
icc_draws |>
  ggplot(aes(x = icc)) +
  geom_density(fill = "steelblue", alpha = 0.4, color = "steelblue4") +
  geom_vline(
    aes(xintercept = mean(icc)),
    linetype = "dashed",
```

```
    color = "steelblue4"
  ) +
  labs(
    title = "Posterior Distribution of ICC",
    subtitle = "Proportion of variance attributable to between-runner differences",
    x = "ICC",
    y = "Density"
  ) +
  theme_bw()
```



25.1.3 Hierarchical model - random intercepts and slopes

```
fit_hier2 <- brm(
  data = running,
  family = gaussian(),
  net ~ 1 + age + (1 + age | runner)
)
```

```
Compiling Stan program...
```

```
Start sampling


SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 1).
Chain 1:
Chain 1: Gradient evaluation took 0.000148 seconds
Chain 1: 1000 transitions using 10 leapfrog steps per transition would take 1.48 seconds.
Chain 1: Adjust your expectations accordingly!
Chain 1:
Chain 1:
Chain 1: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 1: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 1: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 1: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 1: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 1: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 1: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 1: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 1: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 1: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 1: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 1: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 1:
Chain 1:  Elapsed Time: 4.068 seconds (Warm-up)
Chain 1:                3.046 seconds (Sampling)
Chain 1:                7.114 seconds (Total)
Chain 1:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 2).
Chain 2:
Chain 2: Gradient evaluation took 2.9e-05 seconds
Chain 2: 1000 transitions using 10 leapfrog steps per transition would take 0.29 seconds.
Chain 2: Adjust your expectations accordingly!
Chain 2:
Chain 2:
Chain 2: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 2: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 2: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 2: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 2: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 2: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 2: Iteration: 1001 / 2000 [ 50%]  (Sampling)
```

```
Chain 2: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 2: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 2: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 2: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 2: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 2:
Chain 2:  Elapsed Time: 4.089 seconds (Warm-up)
Chain 2:                3.379 seconds (Sampling)
Chain 2:                7.468 seconds (Total)
Chain 2:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 3).
Chain 3:
Chain 3: Gradient evaluation took 2.1e-05 seconds
Chain 3: 1000 transitions using 10 leapfrog steps per transition would take 0.21 seconds.
Chain 3: Adjust your expectations accordingly!
Chain 3:
Chain 3:
Chain 3: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 3: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 3: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 3: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 3: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 3: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 3: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 3: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 3: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 3: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 3: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 3: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 3:
Chain 3:  Elapsed Time: 3.739 seconds (Warm-up)
Chain 3:                3.688 seconds (Sampling)
Chain 3:                7.427 seconds (Total)
Chain 3:

SAMPLING FOR MODEL 'anon_model' NOW (CHAIN 4).
Chain 4:
Chain 4: Gradient evaluation took 2.3e-05 seconds
Chain 4: 1000 transitions using 10 leapfrog steps per transition would take 0.23 seconds.
Chain 4: Adjust your expectations accordingly!
Chain 4:
Chain 4:
```

```
Chain 4: Iteration:    1 / 2000 [  0%]  (Warmup)
Chain 4: Iteration:  200 / 2000 [ 10%]  (Warmup)
Chain 4: Iteration:  400 / 2000 [ 20%]  (Warmup)
Chain 4: Iteration:  600 / 2000 [ 30%]  (Warmup)
Chain 4: Iteration:  800 / 2000 [ 40%]  (Warmup)
Chain 4: Iteration: 1000 / 2000 [ 50%]  (Warmup)
Chain 4: Iteration: 1001 / 2000 [ 50%]  (Sampling)
Chain 4: Iteration: 1200 / 2000 [ 60%]  (Sampling)
Chain 4: Iteration: 1400 / 2000 [ 70%]  (Sampling)
Chain 4: Iteration: 1600 / 2000 [ 80%]  (Sampling)
Chain 4: Iteration: 1800 / 2000 [ 90%]  (Sampling)
Chain 4: Iteration: 2000 / 2000 [100%]  (Sampling)
Chain 4: 
Chain 4:  Elapsed Time: 4.317 seconds (Warm-up)
Chain 4:                3.885 seconds (Sampling)
Chain 4:                8.202 seconds (Total)
Chain 4: 

Warning: There were 146 divergent transitions after warmup. See
https://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
to find out why this is a problem and how to eliminate them.

Warning: Examine the pairs() plot to diagnose sampling problems

Warning: The largest R-hat is 1.05, indicating chains have not mixed.
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#r-hat

Warning: Bulk Effective Samples Size (ESS) is too low, indicating posterior means and medians
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#bulk-ess

Warning: Tail Effective Samples Size (ESS) is too low, indicating posterior variances and ta
Running the chains for more iterations may help. See
https://mc-stan.org/misc/warnings.html#tail-ess
```

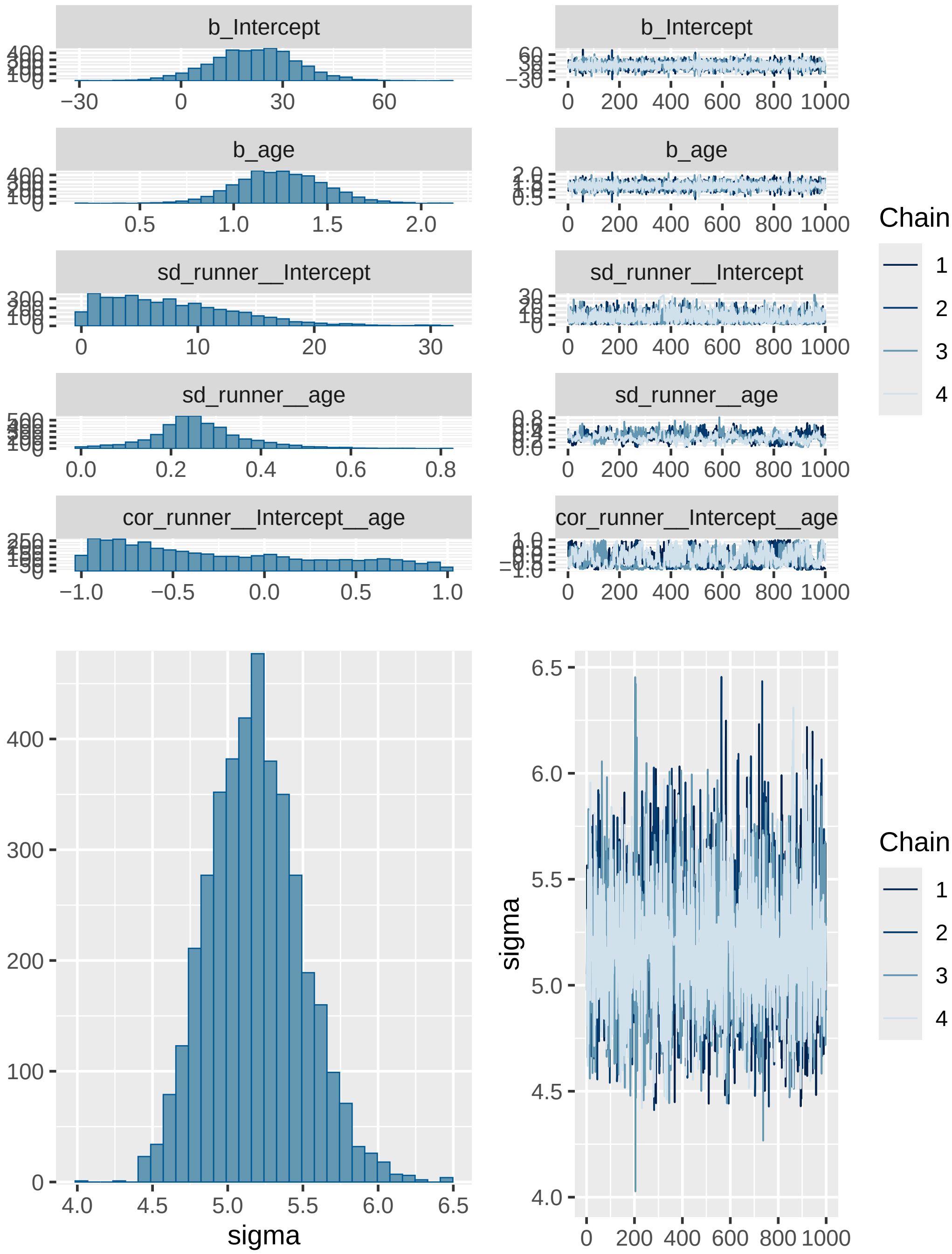
```r
summary(fit_hier2)
```

```
Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations and/or
setting stronger priors.
```

```
Warning: There were 146 divergent transitions after warmup. Increasing
adapt_delta above 0.8 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup
```

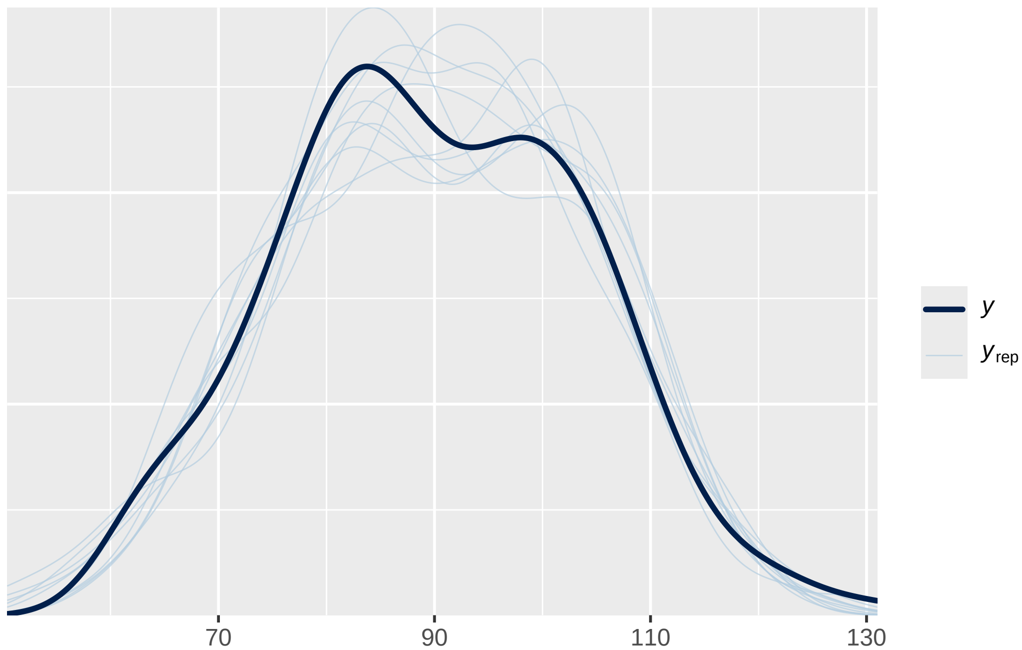
```
 Family: gaussian
  Links: mu = identity
Formula: net ~ 1 + age + (1 + age | runner)
   Data: running (Number of observations: 185)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~runner (Number of levels: 36)
                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)          7.84      5.68     0.34    21.04 1.00     1683     1312
sd(age)                0.26      0.10     0.06     0.50 1.04      165      493
cor(Intercept,age)    -0.24      0.56    -0.97     0.90 1.05      108      313

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    21.44     12.65    -3.61    45.96 1.00     5397     2608
age           1.25      0.23     0.80     1.71 1.00     5157     2750

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     5.18      0.31     4.62     5.81 1.00     3398     2347

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(fit_hier2)
```

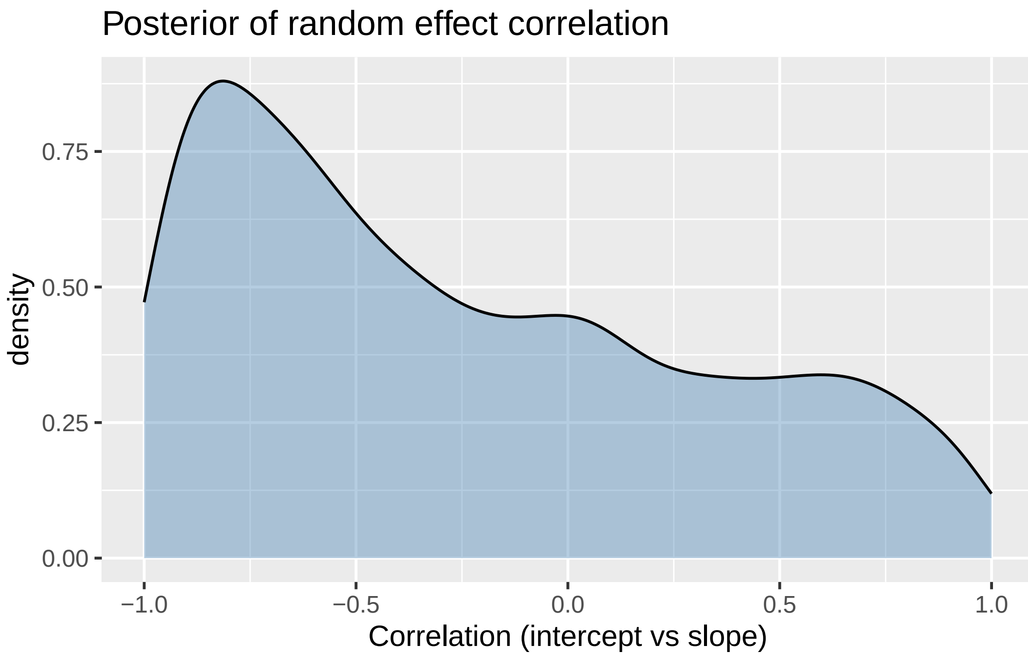



```
pp_check(fit_hier2)
```

```
Using 10 posterior draws for ppc type 'dens_overlay' by default.
```




```
fit_hier2 |>
  spread_draws(cor_runner__Intercept__age) |>
  ggplot(aes(x = cor_runner__Intercept__age)) +
  geom_density(fill = "steelblue", alpha = 0.4) +
  scale_x_continuous(limits = c(-1, 1)) +
  labs(
    x = "Correlation (intercept vs slope)",
    title = "Posterior of random effect correlation"
  )
```




```
library(shinystan)

launch_shinystan(fit_hier2)
```